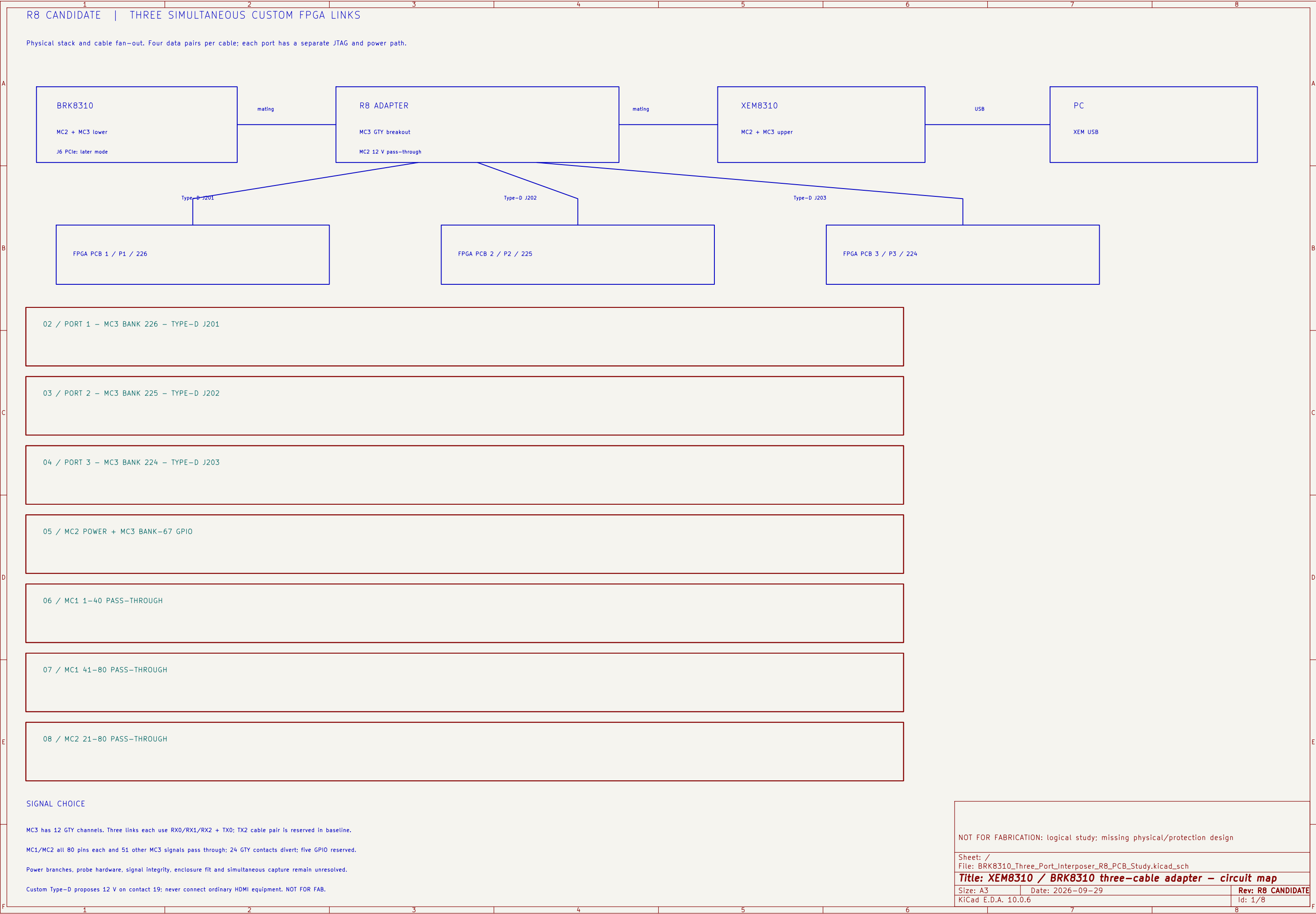
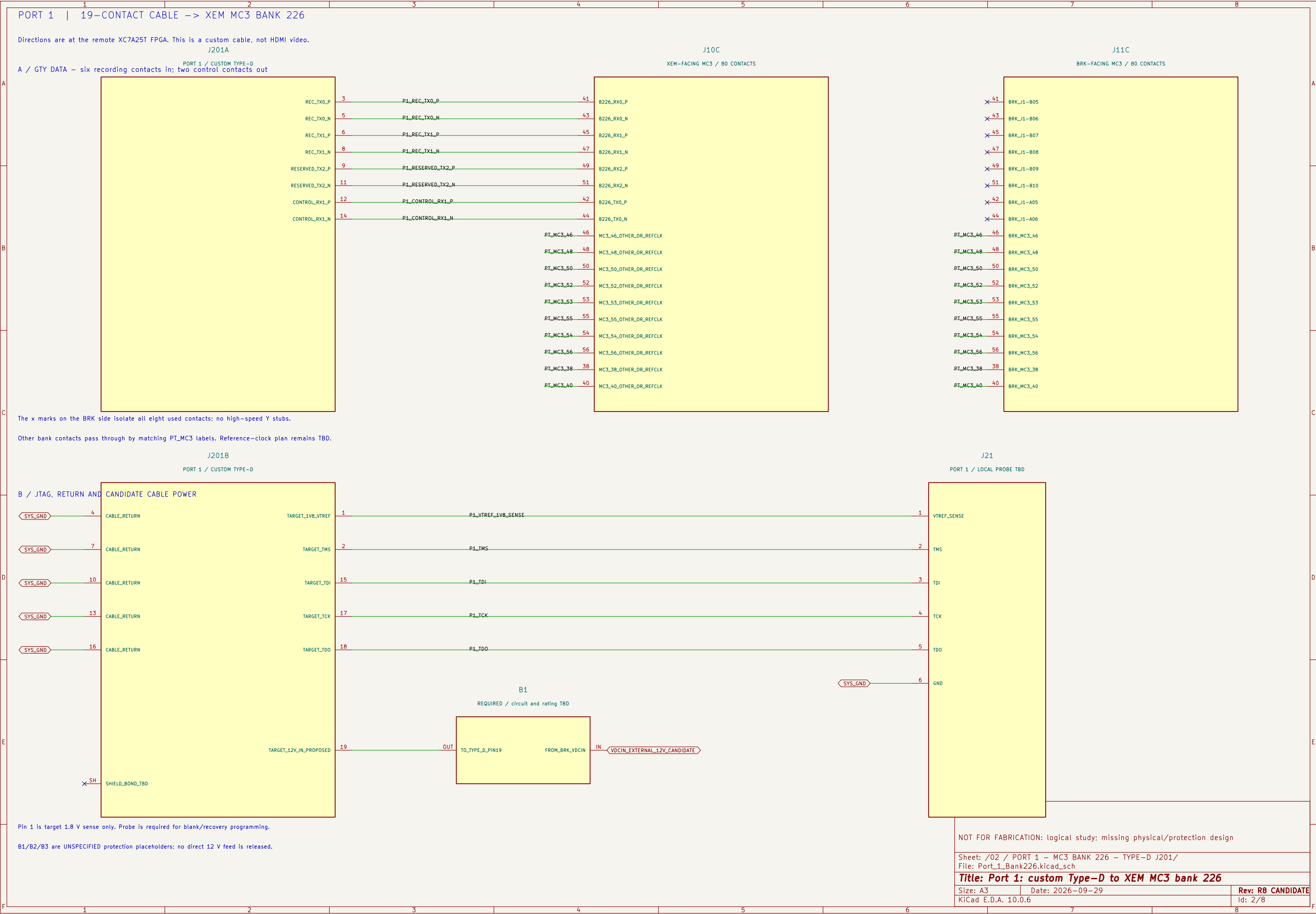
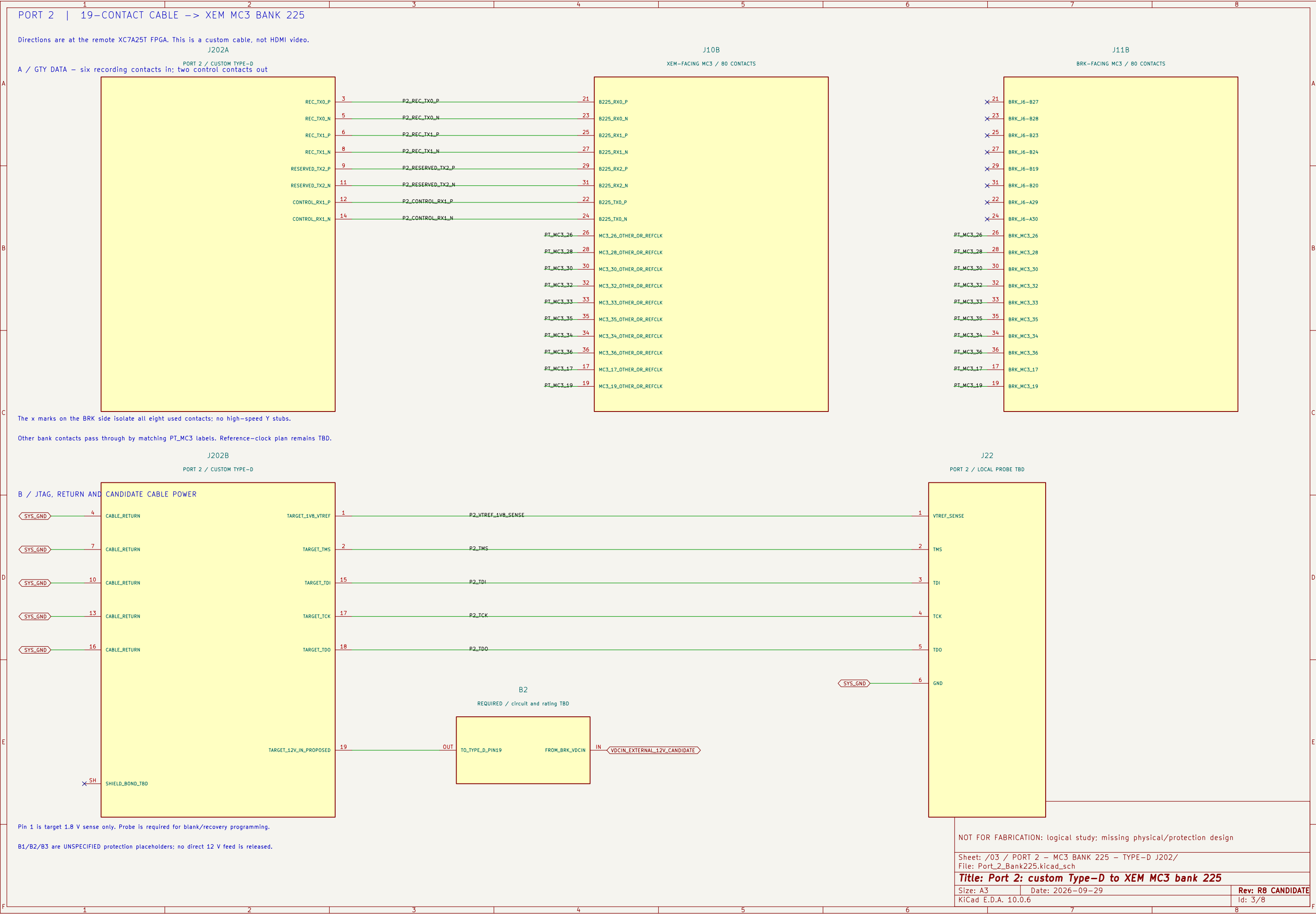
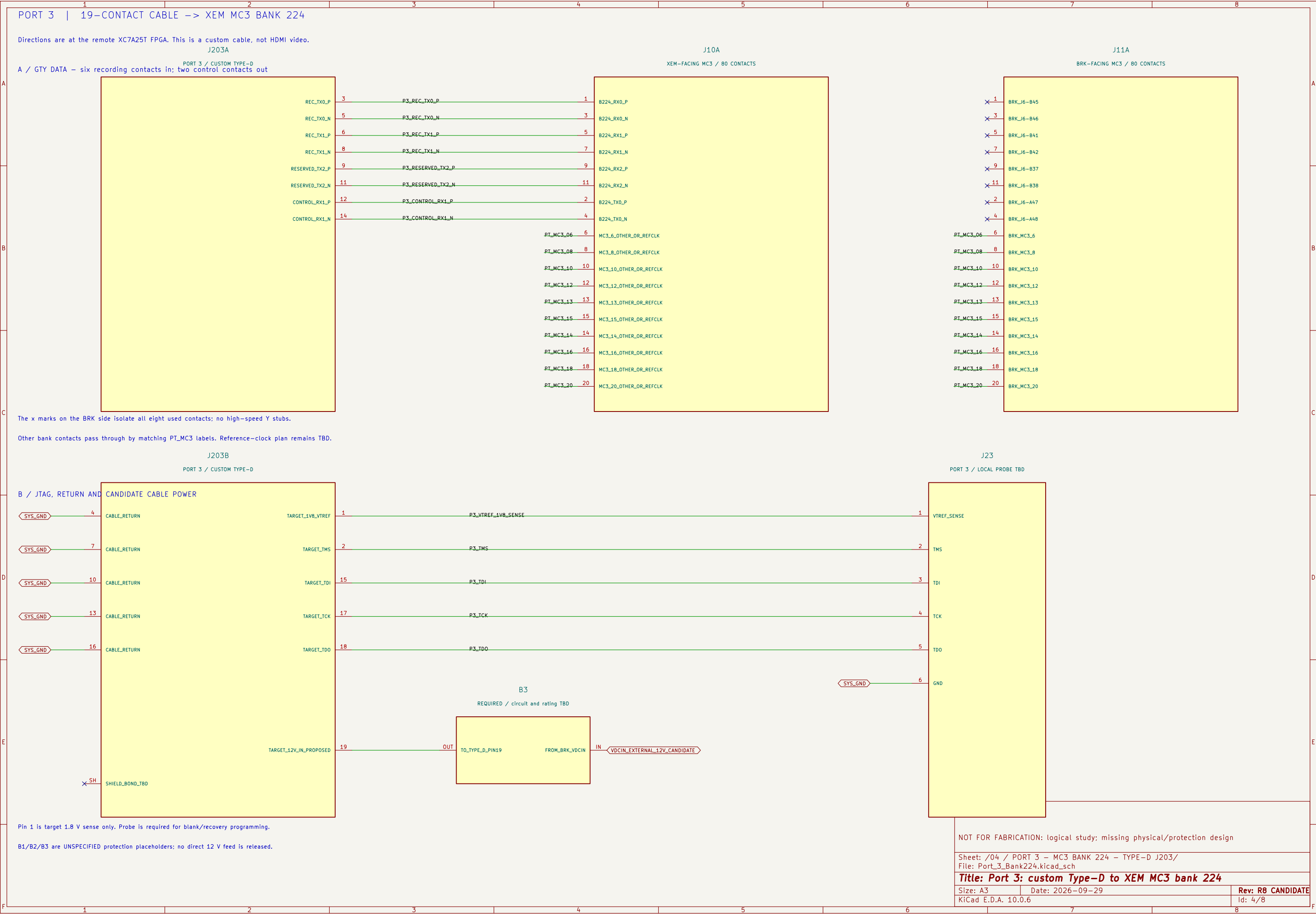










BRK_6-B45

1

BRK_6-B46

3

BRK_6-B41

5

BRK_6-B42

7

BRK_6-B37

9

BRK_6-B38

11

BRK_6-A47

2

BRK_6-A48

4

BRK_MC3_6

6

BRK_MC3_8

8

BRK_MC3_10

10

BRK_MC3_12

12

BRK_MC3_13

13

BRK_MC3_15

15

BRK_MC3_14

14

BRK_MC3_16

16

BRK_MC3_18

18

BRK_MC3_20

20

The x marks on the BRK side isolate all eight used contacts; no high-speed Y stubs.

Other bank contacts pass through by matching PT_MC3 labels. Reference-clock plan remains TBD.

J203B

PORT 3 / CUSTOM TYPE-D

B / JTAG, RETURN AND CANDIDATE CABLE POWER

SYS_GND

4

CABLE_RETURN

TARGET_1V8_VTREF

1

P3_VTREF_1V8_SENSE

1

VTREF_SENSE

SYS_GND

7

CABLE_RETURN

TARGET_TMS

2

P3_TMS

2

TMS

SYS_GND

10

CABLE_RETURN

TARGET_TDI

15

P3_TDI

3

TDI

SYS_GND

13

CABLE_RETURN

TARGET_TCK

17

P3_TCK

4

TCK

SYS_GND

16

CABLE_RETURN

TARGET_TDO

18

P3_TDO

5

TDO

SYS_GND

6

GND

B3

REQUIRED / circuit and rating TBD

TO_TYPE_D_PIN19

FROM_BRK_VDCIN

IN

VDCIN_EXTERNAL_12V_CANDIDATE

OUT

SH

SHIELD_BOND_TBD

NOT FOR FABRICATION: logical study; missing physical/protection design

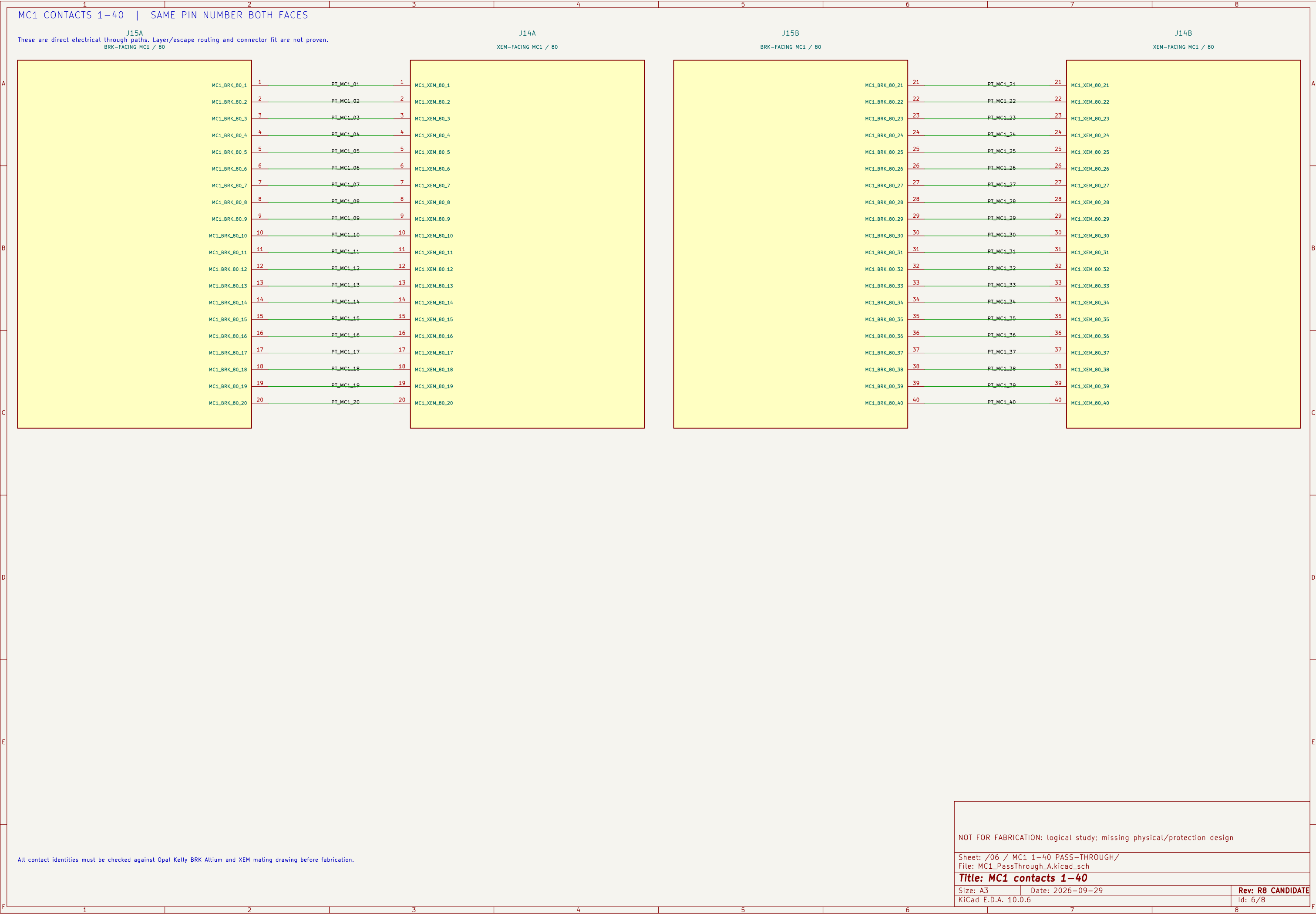
Sheet: /04 / PORT 3 - MC3 BANK 224 - TYPE-D J203/
File: Port_3_Bank224.kicad_sch

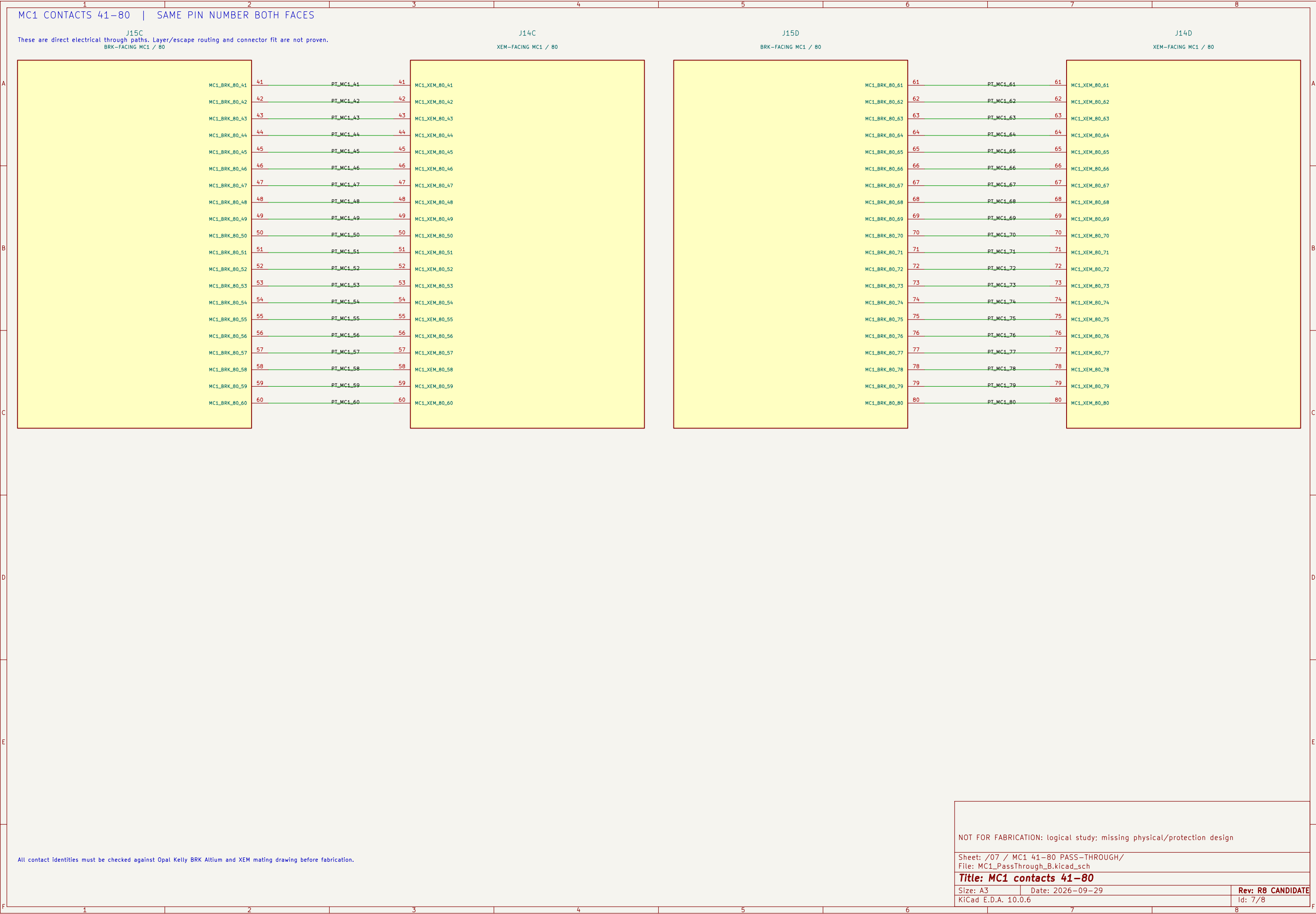
Title: Port 3: custom Type-D to XEM MC3 bank 224

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-29

Rev: R8 CANDIDATE
Id: 4/8





MC2 CONTACTS 21–80 SAME PIN NUMBER BOTH FACES											
J13B These are direct electrical through paths. Layer/escape routing and connector fit are not proven. BRK–FACING MC2 / 80				J12B XEM–FACING MC2 / 80		J13C BRK–FACING MC2 / 80		J12C XEM–FACING MC2 / 80			
MC2_BRK_80_21 21 PT_MC2_21 21 MC2_BRK_80_22 22 PT_MC2_22 22 MC2_BRK_80_23 23 PT_MC2_23 23 MC2_BRK_80_24 24 PT_MC2_24 24 MC2_BRK_80_25 25 PT_MC2_25 25 MC2_BRK_80_26 26 PT_MC2_26 26 MC2_BRK_80_27 27 PT_MC2_27 27 MC2_BRK_80_28 28 PT_MC2_28 28 MC2_BRK_80_29 29 PT_MC2_29 29 MC2_BRK_80_30 30 PT_MC2_30 30 MC2_BRK_80_31 31 PT_MC2_31 31 MC2_BRK_80_32 32 PT_MC2_32 32 MC2_BRK_80_33 33 PT_MC2_33 33 MC2_BRK_80_34 34 PT_MC2_34 34 MC2_BRK_80_35 35 PT_MC2_35 35 MC2_BRK_80_36 36 PT_MC2_36 36 MC2_BRK_80_37 37 PT_MC2_37 37 MC2_BRK_80_38 38 PT_MC2_38 38 MC2_BRK_80_39 39 PT_MC2_39 39 MC2_BRK_80_40 40 PT_MC2_40 40				MC2_XEM_80_21 MC2_XEM_80_22 MC2_XEM_80_23 MC2_XEM_80_24 MC2_XEM_80_25 MC2_XEM_80_26 MC2_XEM_80_27 MC2_XEM_80_28 MC2_XEM_80_29 MC2_XEM_80_30 MC2_XEM_80_31 MC2_XEM_80_32 MC2_XEM_80_33 MC2_XEM_80_34 MC2_XEM_80_35 MC2_XEM_80_36 MC2_XEM_80_37 MC2_XEM_80_38 MC2_XEM_80_39 MC2_XEM_80_40		MC2_BRK_80_41 41 PT_MC2_41 41 MC2_BRK_80_42 42 PT_MC2_42 42 MC2_BRK_80_43 43 PT_MC2_43 43 MC2_BRK_80_44 44 PT_MC2_44 44 MC2_BRK_80_45 45 PT_MC2_45 45 MC2_BRK_80_46 46 PT_MC2_46 46 MC2_BRK_80_47 47 PT_MC2_47 47 MC2_BRK_80_48 48 PT_MC2_48 48 MC2_BRK_80_49 49 PT_MC2_49 49 MC2_BRK_80_50 50 PT_MC2_50 50 MC2_BRK_80_51 51 PT_MC2_51 51 MC2_BRK_80_52 52 PT_MC2_52 52 MC2_BRK_80_53 53 PT_MC2_53 53 MC2_BRK_80_54 54 PT_MC2_54 54 MC2_BRK_80_55 55 PT_MC2_55 55 MC2_BRK_80_56 56 PT_MC2_56 56 MC2_BRK_80_57 57 PT_MC2_57 57 MC2_BRK_80_58 58 PT_MC2_58 58 MC2_BRK_80_59 59 PT_MC2_59 59 MC2_BRK_80_60 60 PT_MC2_60 60				MC2_XEM_80_41 MC2_XEM_80_42 MC2_XEM_80_43 MC2_XEM_80_44 MC2_XEM_80_45 MC2_XEM_80_46 MC2_XEM_80_47 MC2_XEM_80_48 MC2_XEM_80_49 MC2_XEM_80_50 MC2_XEM_80_51 MC2_XEM_80_52 MC2_XEM_80_53 MC2_XEM_80_54 MC2_XEM_80_55 MC2_XEM_80_56 MC2_XEM_80_57 MC2_XEM_80_58 MC2_XEM_80_59 MC2_XEM_80_60	
J13D BRK–FACING MC2 / 80				J12D XEM–FACING MC2 / 80							
MC2_BRK_80_61 61 PT_MC2_61 61 MC2_BRK_80_62 62 PT_MC2_62 62 MC2_BRK_80_63 63 PT_MC2_63 63 MC2_BRK_80_64 64 PT_MC2_64 64 MC2_BRK_80_65 65 PT_MC2_65 65 MC2_BRK_80_66 66 PT_MC2_66 66 MC2_BRK_80_67 67 PT_MC2_67 67 MC2_BRK_80_68 68 PT_MC2_68 68 MC2_BRK_80_69 69 PT_MC2_69 69 MC2_BRK_80_70 70 PT_MC2_70 70 MC2_BRK_80_71 71 PT_MC2_71 71 MC2_BRK_80_72 72 PT_MC2_72 72 MC2_BRK_80_73 73 PT_MC2_73 73 MC2_BRK_80_74 74 PT_MC2_74 74 MC2_BRK_80_75 75 PT_MC2_75 75 MC2_BRK_80_76 76 PT_MC2_76 76 MC2_BRK_80_77 77 PT_MC2_77 77 MC2_BRK_80_78 78 PT_MC2_78 78 MC2_BRK_80_79 79 PT_MC2_79 79 MC2_BRK_80_80 80 PT_MC2_80 80				MC2_XEM_80_61 MC2_XEM_80_62 MC2_XEM_80_63 MC2_XEM_80_64 MC2_XEM_80_65 MC2_XEM_80_66 MC2_XEM_80_67 MC2_XEM_80_68 MC2_XEM_80_69 MC2_XEM_80_70 MC2_XEM_80_71 MC2_XEM_80_72 MC2_XEM_80_73 MC2_XEM_80_74 MC2_XEM_80_75 MC2_XEM_80_76 MC2_XEM_80_77 MC2_XEM_80_78 MC2_XEM_80_79 MC2_XEM_80_80							
All contact identities must be checked against Opal Kelly BRK Altium and XEM mating drawing before fabrication.											
NOT FOR FABRICATION: logical study; missing physical/protection design											
Sheet: /08 / MC2 21–80 PASS–THROUGH/ File: MC2_PassThrough.kicad_sch											
Title: MC2 contacts 21–80											
Size: A3			Date: 2026–09–29				Rev: R8 CANDIDATE				
KiCad E.D.A. 10.0.6							Id: 8/8				